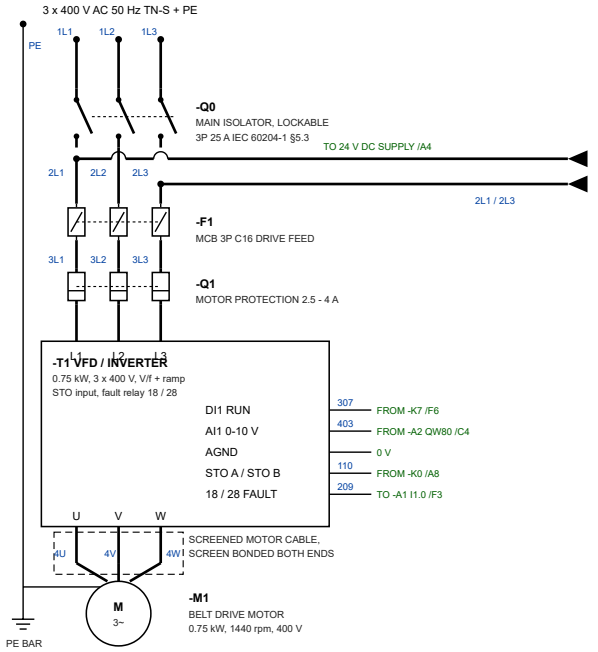
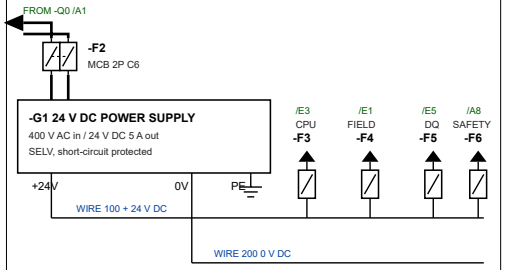
1 POWER SUPPLY AND BELT DRIVE

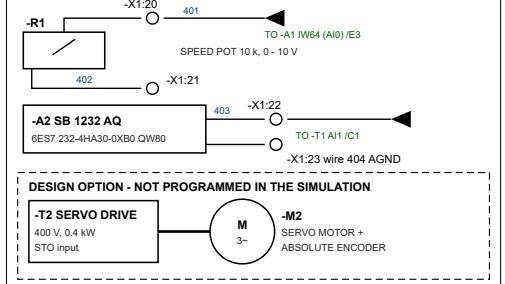


2 24 V DC CONTROL SUPPLY



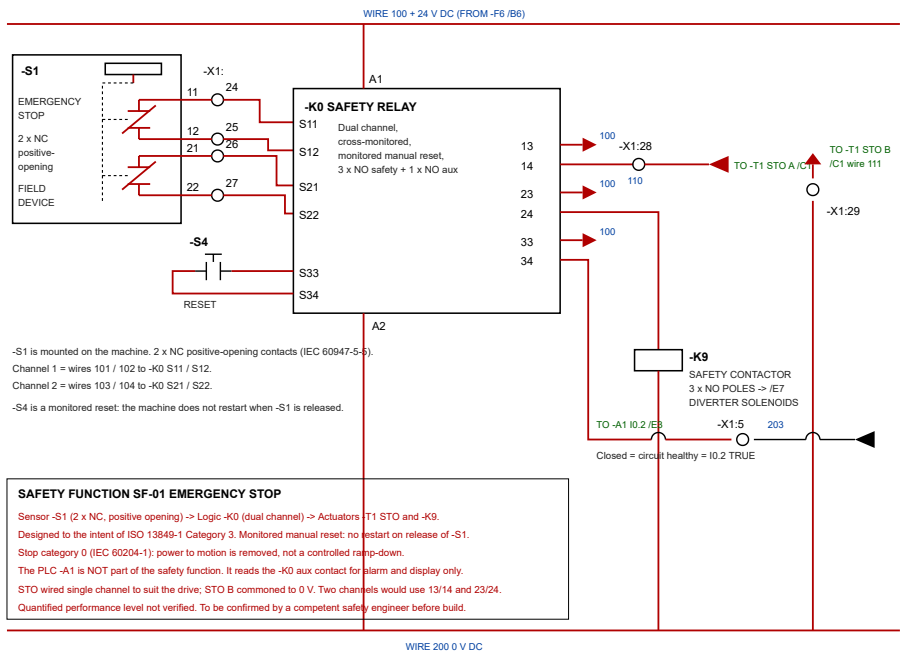
NOTE 3: 0 V is bonded to PE at -G1 only. All control voltage is 24 V DC.

3 ANALOGUE I/O + SERVO TRANSFER (DESIGN OPTION)



Transfer / diverter unit position control. Command via PROFINET or pulse-train.
Detail in electrical_design.md §7. Not implemented in the PLC program.

4 SAFETY CIRCUIT E-STOP / SAFETY RELAY / DRIVE STO



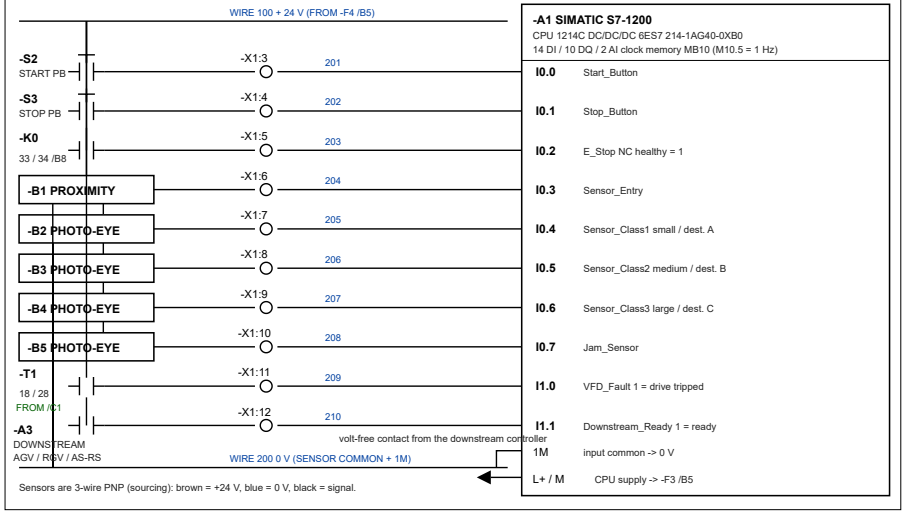
-S1 is mounted on the machine. 2 x NC positive-opening contacts (IEC 60947-5-5).
Channel 1 = wires 101 / 102 to -K0 S11 / S12.
Channel 2 = wires 103 / 104 to -K0 S21 / S22.

-S4 is a monitored reset: the machine does not restart when -S1 is released.

SAFETY FUNCTION SF-01 EMERGENCY STOP

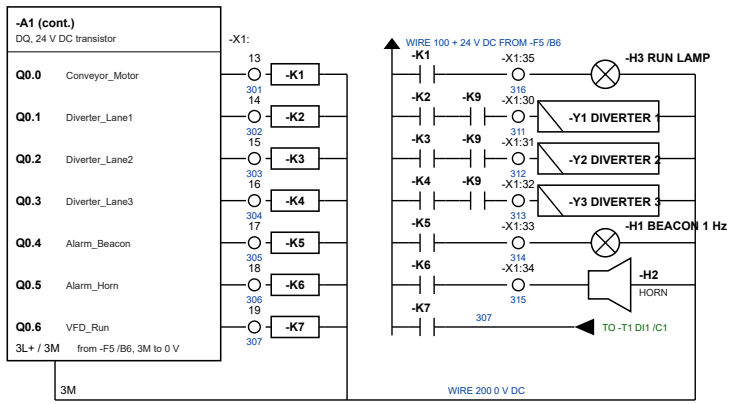
Sensor -S1 (2 x NC, positive opening) -> Logic -K0 (dual channel) -> Actuators -T1 STO and -K9.
Designed to the intent of ISO 13849-1 Category 3. Monitored manual reset: no restart on release of -S1.
Stop category 0 (IEC 60204-1): power to motion is removed, not a controlled ramp-down.
The PLC -A1 is NOT part of the safety function. It reads the -K0 aux contact for alarm and display only.
STO wired single channel to suit the drive; STO B commoned to 0 V. Two channels would use 13/14 and 23/24.
Quantified performance level not verified. To be confirmed by a competent safety engineer before build.

5 -A1 DIGITAL INPUTS I0.0 - I1.1



Sensors are 3-wire PNP (sourcing): brown = +24 V, blue = 0 V, black = signal.

6 -A1 DIGITAL OUTPUTS Q0.0 - Q0.6 AND FIELD LOADS



-K9 poles (/C9) are in series with each solenoid branch, so all three diverters lose power when the safety circuit opens.
Q0.0 is the internal belt-run state (run lamp + line permissive contact). Q0.6 is the run command wired to the inverter.
Interposing relays -K1 to -K7: 24 V DC coil, 6 A contact. The PLC never carries load current directly.

7 NOTES AND LEGEND

NOTES

1. Control wiring 24 V DC, 1.0 mm2. Wire numbers and terminals per the schedule in electrical_design.md.
2. Sensors 3-wire PNP sourcing; 1M common to 0 V.
3. -G1 0 V bonded to PE at one point only.
4. -S1 has 2 x NC positive-opening contacts, wired as two channels into -K0.
5. Safety chain -S1 to -K0 to -T1 STO and -K9. The PLC is not in the safety chain.
6. Stop category 0 for emergency stop. Jam and process faults give a controlled stop by the PLC (category 2).
7. Motor cable screened, screen bonded at both ends on the gland plate.
8. -T2 / -M2 servo transfer unit is a design option, not programmed in the simulation.
9. This sheet is the design basis for EPLAN Electric P8 / AutoCAD Electrical.
10. Field I/O is simulated in S7-PLCSIM. No hardware is installed.

LEGEND

- NO contact — /C1 = cross-reference row C, column 1
- relay / contactor coil — 203 = wire number
- terminal on strip -X1 — -B1 = device tag, IEC 81346-2
- wires crossing, not joined — red = safety circuit

STANDARDS REFERENCED

IEC 60204-1 Electrical equipment of machines - stop functions and categories
ISO 13849-1 Safety-related parts of control systems - category, performance level
IEC 81346-2 Reference designations (-A, -B, -F, -G, -K, -M, -Q, -R, -S, -T, -X, -Y)
IEC 60617 Graphical symbols for diagrams
IEC 61131-3 PLC programming languages - LAD and SCL used in this project
UL 508A / CE Panel and machinery marking requirements considered for export

REVISIONS

REV	DATE	DESCRIPTION	BY
0	2026-08-30	First issue - power, safety circuit, PLC I/O	PSM
1	2026-09-19	Added I1.0 VFD fault, I1.1 downstream ready, Q0.6 VFD run, -K9 solenoid contactor, servo transfer option -T2 / -M2	PSM

IIoT SMART SORTATION CONVEYOR - MATERIAL HANDLING SYSTEM

WAREHOUSE SORTATION CONVEYOR FEEDING A DOWNSTREAM AGV / RGV / AS-RS - SIMULATION-ONLY DESIGN DELIVERABLE

DRAWING TITLE: CONTROL SCHEMATIC - POWER, SAFETY CIRCUIT, PLC I/O		DRAWING No.: PSM-MHS-001	SHEET / REV / SIZE: 1 of 1 Rev 1 A3
DESIGNED BY: PRANAV S M	DATE: 2026-09-19	CHECKED: -	SCALE: NTS
STANDARDS: IEC 60204-1 / ISO 13849-1 / IEC 81346-2 / IEC 60617		INTENDED CAD TOOL: EPLAN Electric P8 / AutoCAD Electrical - this sheet is the design basis	